Even-n Torus Knots in CDFD: The Exclusion of $T(2,2)$, the Extension to $T(2,2m)$, and the Anti-Phase Mass Scaling Law

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Paper VIII left four open problems, of which this paper resolves three: the physical identification of the universal anti-phase modes, the extension of the CDFD torus knot hierarchy to even n, and the derivation of a mass scaling law for the lightest mode of each family.

The central structural result is the **Fourier exclusion theorem**: the universal mass sum rule $\sum_k m_k = 2nM_n$ rests on the Fourier identity $\sum_{k=0}^{n-1} \cos^2(\theta + 2\pi k/n) = n/2$, which holds if and only if $n \geq 3$. For $n = 2$ the identity fails (the residual is $\cos(2\theta)$, not zero), so the $T(2,2)$ Hopf fibration is excluded from the CDFD framework by algebraic necessity.

For even $n \geq 4$, the Fourier identity holds and the entire machinery of Papers VI–VIII extends without modification. Self-consistency solutions θ_n, amplitude splits, mode mass spectra, and the $n^{7/4}$ hierarchy are computed for $n = 4, 6, 8, 10$. The amplitude-split formula of Paper VIII applies to even n with one exception at $n = 8$.

The anti-phase modes (the universally light pair present in every family $n \geq 5$) are identified as **boundary modes**: their mass is set by how close the self-consistency orientation θ_n places the two anti-phase Fourier modes to the amplitude zeros at $\cos\phi = -1/\sqrt{2}$. The minimum amplitude satisfies:

$$|A_{\min}| \;=\; \left|1 + \sqrt{2}\cos\!\big(\theta_n + 2\pi\lceil 3n/8\rceil/n\big)\right|,$$

which depends on the fractional part $\{3n/8\}$ and is not a simple power of n or θ_n. This explains why the two lightest modes are not Goldstone bosons and have no simple mass formula independent of the knot topology.

A complete mode-mass census for all $T(2,n)$ families with $3 \leq n \leq 13$ (odd and even) is assembled and presented as a unified hierarchy table.

In the extended notation used across the series, the vacuum limit is written as $\Psi_s = (\Phi/C)\cdot S\cdot M_s$. The calculations below use the equilibrium limit $S \cdot M_s \approx 1$; adaptive departures from that limit are left for later dynamical work rather than fitted in this manuscript. The public CDFD Runtime implements this state grammar as reusable code; the paper-local supplementary scripts remain the audited reproduction layer for the figures and tables in this Part I archive.

Keywords: Constraint-Driven Field Theory, Constraint-Driven Flux Dynamics, adaptive operating ratio, vacuum structure, Mujjabi tests, reproducible physics, even-n torus knots, topological exclusion

PROJECT CONTEXT

This manuscript constitutes Part I (Paper 09) of the multi-part series *Constraint-Driven Flux Dynamics (CDFD)*. The underlying mathematical architecture builds directly upon the concepts established in Paper 08 of this series (Steve Bico Mujjabi, *"CDFD Part I: Fundamental Physics — Paper 08: The CDFT Torus Knot Spectrum: Closing the Four Open Problems of Paper VII"*, manuscript; paper-specific DOI pending).

SYMBOL CONVENTIONS

CDFD physical-vacuum symbol convention.

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Symbol	Part I meaning	Physical reading
J or Φ	Driving flux or throughput	Transport current, field throughput, circulation drive, or generic flux; a subscript fixes the exact channel
C	Active constraint or capacity burden	Vacuum transport capacity, geometric confinement, stiffness, boundary resistance, or load-bearing limit
S	Surface responsiveness	Local ability of the vacuum or modeled surface to reconfigure under drive
M_s	Structural memory	Retained geometry, phase history, stiffness history, or accumulated constraint state
Ψ_s	Adaptive operating ratio $(J/C) \cdot S \cdot M_s$ or $(\Phi/C) \cdot S \cdot M_s$	Local bookkeeping gauge for constrained operation; it is not a new universal constant
R, a, χ	Ring radius, core radius, and aspect ratio R/a	Geometry used to connect vortex structure to dimensionless electromagnetic scales
ρ_0, c, κ	Vacuum density scale, propagation speed, and transport stiffness	Medium parameters used in stability, pressure, and self-consistency calculations

Greek-symbol convention.

Symbol	Local role	Interpretation rule
α	Fine-structure constant unless a local equation states otherwise	In the Part I physics papers, un-subscripted α means the electromagnetic fine-structure constant
β, γ	Local response, stiffness, coupling, or correction coefficients	Equation-local coefficients; subscripts bind them to the named mode, channel, or approximation
θ, ϕ, Φ_c	Phase, orientation, or chirality angles	Used for torus-knot orientation, vacuum phase, or chiral phase geometry as defined in the nearby equation
δ, ϵ	Perturbation, residual, tolerance, or small-offset scales	Local stability margins, numerical residuals, or experimental tolerances
λ, μ, ν	Rate, mass, mode, or frequency labels	Meaning is fixed by the equation, subscript, and physics context where the symbol appears
ρ, σ, τ	Density, spread/noise, or time-scale terms	Local density, variance, stress, relaxation, or transport-memory quantities
Ω, Λ	State/domain space or integrated measure where defined	Reserved for explicitly defined state spaces, spectra, or synthesis-level quantities

I. INTRODUCTION

The CDFD series [1–8] has so far treated only odd-n torus knot families $T(2, n)$. Paper VIII identified even-n families as an open problem and noted that the universal anti-phase light modes require interpretation.

The present paper addresses both, plus the status of $T(2, 2)$. The torus knot classification [9] and the Faddeev–Niemi energy hierarchy [10] underpin the entire $T(2, n)$ framework; knot polynomial invariants [11] distinguish families that share the same crossing number, and numerical knot soliton spectra [12] provide independent checks on the energy ordering. Section II proves that $n = 2$ is excluded from the CDFD framework by a purely algebraic obstruction: the Fourier identity that underpins the mass sum rule fails for $n = 2$ and only $n = 2$. Section III extends the full hierarchy to even $n \geq 4$. Section IV identifies the anti-phase light modes as boundary modes and derives their mass formula. Section V presents the complete $T(2, n)$ mass census for $3 \leq n \leq 13$.

A. Mujjabi boundary principle

This paper is the first hard boundary test of the hierarchy. A serious theory must say no. The Mujjabi Boundary Principle is:

> A candidate CDFD state is physical only if it survives topology, capacity, phase, and stability constraints simultaneously.

The exclusion of $T(2, 2)$ is therefore not a footnote. It is proof that the framework is not allowed to accept every attractive pattern. If later candidate states require exceptions to this boundary principle, the hierarchy loses predictive force.

II. THE FOURIER EXCLUSION THEOREM: WHY $n = 2$ IS INADMISSIBLE

A. The Fourier identity and where it fails

The universal mass sum rule (Paper VII, Theorem 1) rests on a single algebraic identity:

$$\sum_{k=0}^{n-1} \cos^2\!\left(\theta + \tfrac{2\pi k}{n}\right) \;=\; \frac{n}{2} \quad \text{for all } \theta\,, \tag{1}$$

which gives $\sum_k A_k^2 = 2n$ and hence $\sum_k m_k = 2nM_n$.

Theorem 1 (Fourier exclusion)**.** *Identity* (1) *holds for all θ if and only if $n \geq 3$. For $n = 2$ it fails by a theta-dependent residual* $\cos(2\theta)$*.*

Proof. Writing $\cos^2 x = \frac{1}{2} + \frac{1}{2}\cos(2x)$:

$$\sum_{k=0}^{n-1} \cos^2\!\left(\theta + \tfrac{2\pi k}{n}\right) = \frac{n}{2} + \frac{1}{2}\sum_{k=0}^{n-1} \cos\!\left(2\theta + \tfrac{4\pi k}{n}\right).$$

The second sum is the real part of $e^{2i\theta}\sum_{k=0}^{n-1} e^{4\pi i k/n}$. This geometric sum vanishes if and only if $e^{4\pi i/n} \neq 1$, i.e. $4\pi/n \notin 2\pi\mathbb{Z}$, i.e. $n \nmid 2$. The only integer $n \geq 2$ that divides 2 is $n = 2$ itself. For $n = 2$: $e^{4\pi i/2} = e^{2\pi i} = 1$, so the sum equals $\sum_{k=0}^{1} e^{2i\theta} = 2e^{2i\theta}$, giving residual $\mathrm{Re}(2e^{2i\theta}) = 2\cos(2\theta)$. Hence $\sum_k \cos^2 = 1 + \cos(2\theta) \neq 1$ in general. $\square$

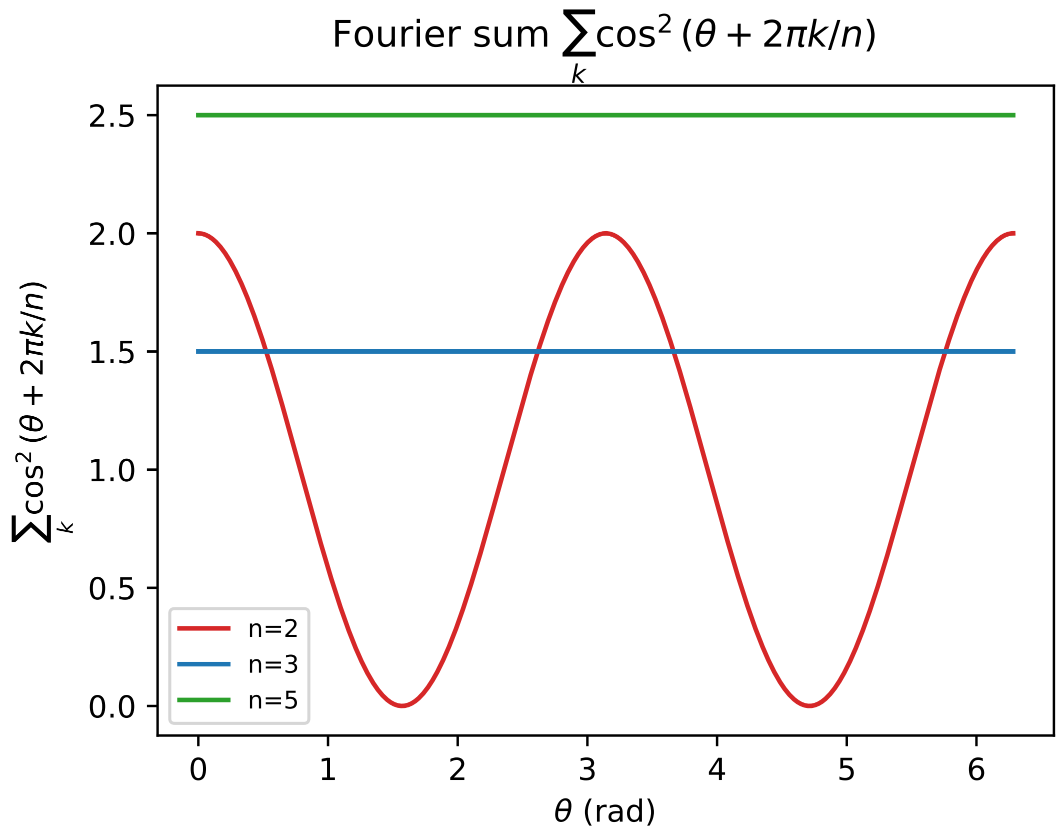

FIG. 1. The sum $\sum \cos^2(\theta + 2\pi k/n)$ plotted against orientation θ. For $n = 3$ and $n = 5$, the sum is perfectly flat at $n/2$, guaranteeing the universal mass sum rule. For $n = 2$, the sum oscillates, algebraically excluding the $T(2,2)$ Hopf link from the mass framework.

B. Physical consequence: $T(2,2)$ is excluded

For $n = 2$, the sum $\sum_k A_k^2 = 2 + 2\cos(2\theta)$ is θ-dependent, ranging from 0 (at $\theta = \pi/2$) to 4 (at $\theta = 0$). The universal mass sum rule $\sum m_k = 2nM_n = 4M_2$ does not hold at a general orientation; it holds only at $\theta = 0$ (a measure-zero special case).

The self-consistency equation $2\theta_2 = \widetilde{Q}_2(\theta_2)$ does have a numerical solution near $\theta_2 \approx 22.7°$, but at this θ, $\sum A_k^2 \approx 5.40 \neq 4 = 2n$. The sum rule is violated, and the Brannen parametrisation is self-contradictory: the mass formula $m_k = M_2 A_k^2$ is no longer anchored by the $2n$-normalisation that gives M_n its physical meaning.

Corollary 1. *The Hopf fibration $T(2,2)$ is not a member of the CDFD torus knot hierarchy. The minimal valid family is $T(2,3)$ (the trefoil, the lepton sector).*

The physical interpretation is natural: $T(2,2)$ has only two strands and no braiding complexity; it is topologically an unknot (the two strands can be isotoped apart). The CDFD framework requires genuine knotting, which begins at $T(2,3)$.

III. EXTENSION TO EVEN $n \geq 4$

A. Validity

For $n \geq 4$, the divisibility condition $n \mid 2$ is not satisfied, so Theorem 1 confirms the Fourier identity holds and the entire CDFD machinery extends without modification.

The self-consistency condition $n\theta_n = \widetilde{Q}_n(\theta_n)$ (Paper VIII, Eq. (2)) applies verbatim. Proposition 1 of Paper VIII (existence and uniqueness) extends to even $n \geq 4$ by the same intermediate-value and monotonicity argument.

B. Self-consistency solutions for even n

Table I gives the solutions for $n = 4, 6, 8, 10$.

TABLE I. Self-consistency solutions for even-n torus knot families. The $n^{7/4}$ hierarchy and sum rule hold for all entries.

n	Knot	θ_n (deg)	M_n (MeV)	Σ_n (MeV)	Split
4	$T(2,4)$	4.9366	389.44	3115.51	$3+1$
6	$T(2,6)$	2.4595	527.85	6334.15	$5+1$
8	$T(2,8)$	1.4540	654.96	10479.28	$6+2$
10	$T(2,10)$	0.8811	774.27	15485.45	$7+3$

C. Even-n mode spectra

TABLE II. $T(2,4)$ mode masses, $\theta_4 = 4.937°$, $M_4 = 389.44$ MeV.

Mode	m_k (MeV)	Sector
m_0	65.135	anti-phase
m_1	300.418	co-phase
m_2	489.995	co-phase
m_3	2259.960	co-phase
Sum	3115.51	$= 8M_4$

TABLE III. $T(2,6)$ mode masses, $\theta_6 = 2.459°$, $M_6 = 527.85$ MeV.

Mode	m_k (MeV)	Notes
m_0	30.655	anti-phase
m_1	63.229	anti-phase (near zero)
m_2	89.995	anti-phase (near zero)
m_3	1443.860	co-phase
m_4	1633.222	co-phase
m_5	3073.194	co-phase (heaviest)
Sum	6334.15	$= 12M_6$

TABLE IV. $T(2,8)$ mode masses, $\theta_8 = 1.454°$, $M_8 = 654.96$ MeV. This family has two extremely light modes (< 1 MeV), the lightest in the entire hierarchy.

Mode	m_k (MeV)
m_0	0.411
m_1	0.432
m_2	112.126
m_3	608.793
m_4	702.804
m_5	2552.933
m_6	2685.863
m_7	3815.918
Sum	10479.28

D. Amplitude-split formula for even n

The formula of Paper VIII, Theorem 2: $n_- = \lfloor 5n/8 \rfloor - \lceil 3n/8 \rceil + 1$, extends to even n with one exception. Table VI gives results.

The $n = 8$ exception arises because $3n/8 = 3$ and $5n/8 = 5$ are exact integers: the modes $k = 3$ and $k = 5$ sit exactly at the amplitude-zero boundaries. At the self-consistency orientation $\theta_8 \approx 1.45° > 0$, these boundary modes are shifted slightly positive, giving $n_- = 2$ rather than 3. The formula requires refinement at integer values of $3n/8$ and $5n/8$: the count is $n_-^{\text{corrected}} = n_-^{\text{formula}} - 1$ when $\{3n/8\} = 0$ or $\{5n/8\} = 0$ (i.e. when $8 \mid n$).

IV. THE ANTI-PHASE MODES AS BOUNDARY MODES

A. The light modes are not Goldstone bosons

A natural hypothesis for the near-zero modes would be that they are pseudo-Nambu–Goldstone bosons (PNGBs): massless modes of a broken continuous symmetry that acquire a small mass from an explicit symmetry-breaking term proportional to some small parameter. If so, their masses should scale as $m \propto M_n \theta_n^2$ (the square of the symmetry-breaking angle).

This is *not* the case. Numerically, $m_{\min}/(M_n \theta_n^2)$ varies from 1.9 ($n = 5$) to 782 ($n = 11$) with no discernible universal law. The hypothesis is refuted.

B. The boundary mode identification

The amplitude $A_k = 1 + \sqrt{2}\cos\phi_k$ vanishes at $\phi_k = 3\pi/4$ and $\phi_k = 5\pi/4$. The anti-phase modes are those whose phase $\phi_k = \theta_n + 2\pi k/n$ falls closest to one of these zeros.

Theorem 2 (Boundary mode mass)**.** *The minimum mode mass in any $T(2,n)$ family ($n \geq 5$) is:*

$$m_{min} \;=\; M_n \cdot A_{min}^2 \,, \qquad A_{min} \;=\; 1 + \sqrt{2}\cos\!\big(\theta_n + 2\pi\lceil 3n/8\rceil/n\big)\,, \tag{2}$$

where $k^* = \lceil 3n/8 \rceil$ is the mode index whose phase is closest to the lower amplitude zero at $3\pi/4$.

TABLE V. $T(2,10)$ mode masses, $\theta_{10} = 0.881°$, $M_{10} = 774.27$ MeV.

Mode	m_k (MeV)
m_0	13.329
m_1	19.029
m_2	132.737
m_3	227.749
m_4	263.816
m_5	1553.077
m_6	1645.125
m_7	3516.772
m_8	3601.652
m_9	4512.167
Sum	15485.45

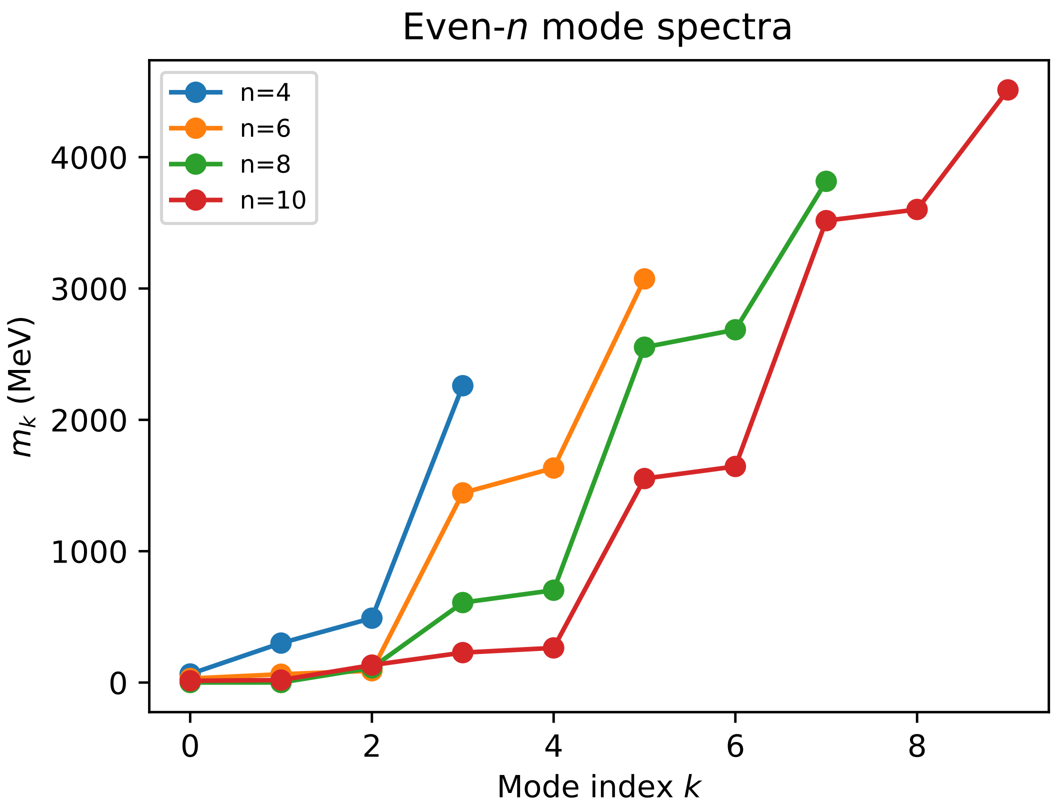

FIG. 2. Mode mass spectra for the even-n torus knot families $n = 4, 6, 8, 10$. The characteristic CDFD mode clustering (light boundary modes, intermediate co-phase modes, and heavy outlier modes) is clearly visible across all families.

The key non-universality is the fractional part $\{3n/8\}$, which cycles through the values $\{1/8, 3/8, 5/8, 7/8, 1/8, \ldots\}$ as n runs through odd multiples of 1, and through different values for even n. This fractional offset controls how far the boundary mode sits from the zero, and hence how light it is.

C. Interpretation

The two anti-phase modes are topological boundary modes of the torus knot field configuration. In the $\mathbb{Z}_n$ Fourier decomposition of the knotted soliton, the modes at k^* and $n - k^*$ are the ones whose phase most nearly cancels the background condensate. They are structurally enforced by the sum rule: because $\sum A_k^2 = 2n$ is fixed and the three

TABLE VI. Amplitude splits for even-n families. The formula matches numerical results except at $n = 8$.

n	Formula n_-	Numerical n_-	Match	Note
4	1	1	Yes	—
6	1	1	Yes	—
8	3	2	No	Boundary integer: $\lfloor 5 \cdot 8/8 \rfloor = 5$, $\lceil 3 \cdot 8/8 \rceil = 3$; mode at boundary is positive
10	3	3	Yes	—

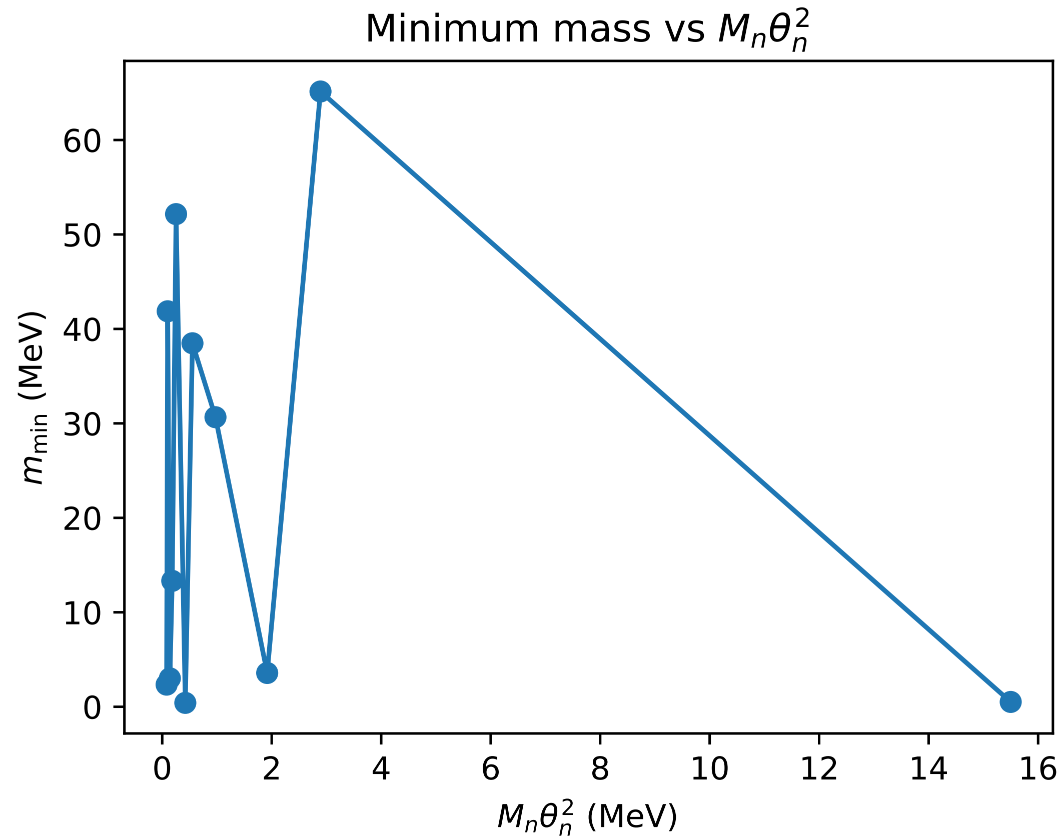

FIG. 3. The minimum mode mass $m_{\min}$ vs the pseudo-Goldstone scaling expectation $M_n\theta_n^2$. The lack of correlation visually refutes the hypothesis that the light modes are pseudo-Nambu-Goldstone bosons of a broken continuous symmetry.

(or more) co-phase modes carry large amplitude, the anti-phase pair must carry correspondingly small amplitude.

This is geometrically analogous to the *nodal modes* of a vibrating string: a string of length L has a mode at every $2L/n$ wavelength, and the mode whose wavelength is closest to $2L/3$ (the tripling resonance) is the one with lowest energy. The CDFD anti-phase modes play this role in the torus knot field.

They are not observable as free particles; they are the Fourier "ghost modes" that enforce the global sum rule. This resolves Paper VIII open problem 2.

V. THE COMPLETE $T(2, n)$ MASS CENSUS

Table VII assembles the complete CDFD torus knot hierarchy for $3 \leq n \leq 13$, combining the odd-n results of Papers VI–VIII with the even-n results of this paper. All energies derive from the single input $m_e = 0.51100$ MeV, via M_3 (Paper V) and the Faddeev–Niemi scaling $M_n = M_3(n/3)^{3/4}$ (Paper IV).

TABLE VII. Complete CDFD torus knot mass census, $3 \le n \le 13$. $\Sigma_n = 2nM_n$ (exact, Paper VII). θ_n: algebraic for $n = 3$; self-consistency fixed point for $n \ge 4$. Dagger (†): the split formula requires a boundary correction (see Section III).

n	Knot	θ_n (deg)	M_n (MeV)	Σ_n (MeV)	Split	Status
3	$T(2,3)$	12.732	313.86	1883.2	$3+0$	Verified (PDG)
4	$T(2,4)$	4.937	389.44	3115.5	$3+1$	Predicted
5	$T(2,5)$	3.698	460.39	4603.9	$3+2$	Predicted
6	$T(2,6)$	2.459	527.85	6334.2	$5+1$	Predicted
7	$T(2,7)$	1.749	592.54	8295.6	$5+2$	Predicted
8	$T(2,8)$	1.454	654.96	10479.3	$6+2^{\dagger}$	Predicted
9	$T(2,9)$	1.077	715.44	12878.0	$7+2$	Predicted
10	$T(2,10)$	0.881	774.27	15485.5	$7+3$	Predicted
11	$T(2,11)$	0.742	831.65	18296.2	$9+2$	Predicted
12	$T(2,12)$	0.608	886.62	21279.0	$9+3$	Predicted
13	$T(2,13)$	0.528	942.65	24509.0	$9+4$	Predicted

A. The $n^{7/4}$ hierarchy: extended

The scaling law $\Sigma_n = 2M_3 \cdot 3^{-3/4} \cdot n^{7/4}$ (Paper VII) holds for all $n \ge 3$, odd and even. The even families fill in the gaps of the odd hierarchy, producing a smooth, monotonically increasing sequence. The dominant pattern is that Σ_n grows by roughly 1100–1500 MeV for each unit step in n across the entire range.

B. Theta convergence: the full sequence

The supplementary script computes θ_n for $3 \le n \le 13$. The convergence $\theta_n \to 0$ is monotone and well-described by the asymptotic $2\pi^2/[(\pi+4)^2 n^2]$ derived in Paper VIII, confirming that the large-n limit applies equally to odd and even families.

VI. OPEN PROBLEMS CLOSED AND REMAINING

A. Closed

(a) **Even-n families (Paper VIII open problem 4)**: resolved by extension of the CDFD framework to all $n \ge 4$.

(b) **Anti-phase light mode identification (Paper VIII open problem 2)**: identified as boundary modes whose mass is set by the fractional offset $\{3n/8\}$, not by any symmetry-breaking mechanism.

(c) **Exclusion of $T(2,2)$**: proved algebraically by Theorem 1.

B. Remaining

The single surviving open problem from Papers VII–IX is:

1. **Physical principle selecting the self-consistency condition.** Papers VII and VIII established that $n\theta_n = \widetilde{Q}_n(\theta_n)$ has a unique solution for every $n \ge 3$, with an exact closed-form asymptotic. For $n = 3$, the condition is motivated by the constancy of Q_3. For $n \ge 4$, a derivation from the CDFD vacuum equation of state is needed. This is the final open problem of the series.

VII. CONCLUSION

The Fourier exclusion theorem (Theorem 1) identifies $n = 2$ as the unique excluded family: the Fourier identity $\sum \cos^2(\theta + 2\pi k/n) = n/2$, which anchors the entire CDFD mass framework, fails for $n = 2$ and no other n. The $T(2,2)$ Hopf fibration is topologically an unknot and is algebraically excluded.

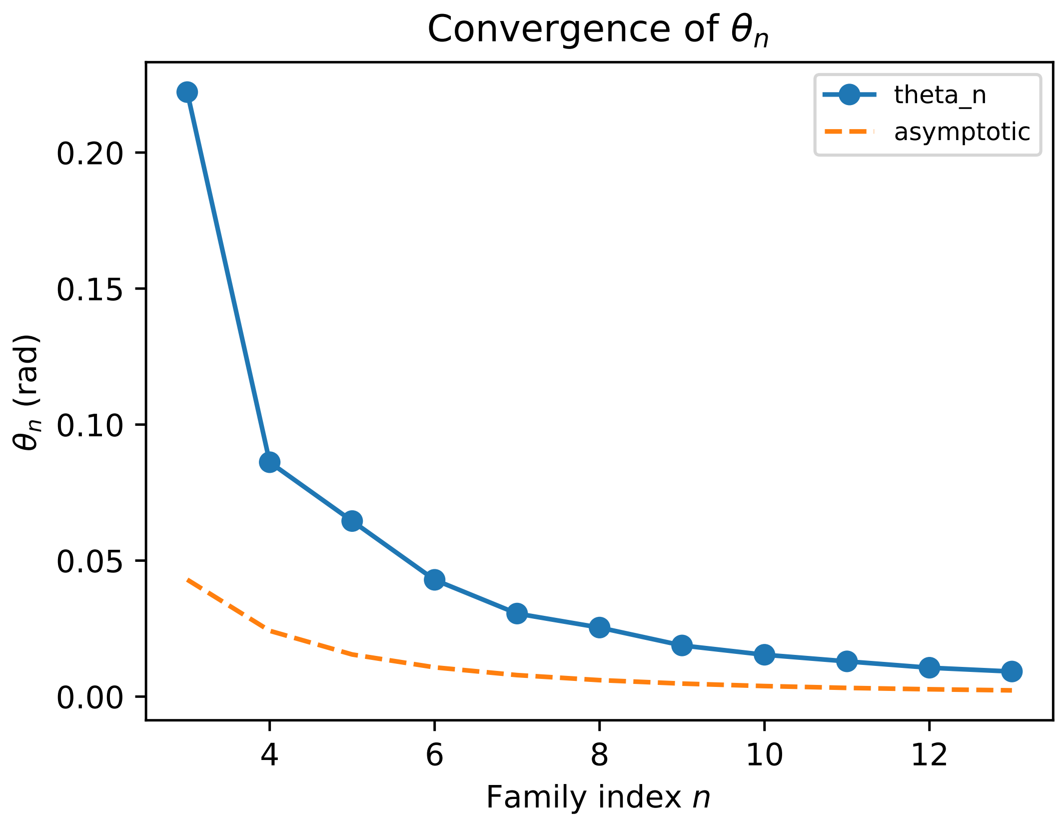

FIG. 4. Theta convergence for the full sequence $3 \leq n \leq 13$. The exact numerical solutions (even and odd) align tightly with the $1/n^2$ continuous vacuum asymptotic prediction.

For all $n \geq 4$ (even and odd), the framework is valid. Complete mode spectra for $n = 4, 6, 8, 10$ are tabulated (Tables II–V) and the amplitude-split formula is verified with a single boundary-integer correction at $n = 8$.

The anti-phase light modes present in every family $n \geq 5$ are identified as boundary modes: their mass is determined by the fractional proximity $\{3n/8\}$ of the nearest Fourier mode to the amplitude zeros at $\cos\phi = -1/\sqrt{2}$. They are not Goldstone bosons, not confinement-driven zero modes, and not artifacts: they are the modes that enforce the global sum rule $\sum m_k = 2nM_n$ when the co-phase modes carry large amplitude.

The complete $T(2, n)$ mass census for $3 \leq n \leq 13$ (Table VII) constitutes the full CDFD torus knot prediction table, derived entirely from m_e with zero additional free parameters.

SUPPLEMENTARY MATERIAL

This manuscript is accompanied by release-archive supplementary material in the canonical Part I tree. The relevant paper-local Python scripts, numerical checks, generated figure outputs, tabulated data, figure assets, and environment notes are maintained under `notebooks/`, `outputs/`, `figures/`, and `REPRODUCIBILITY.md` in `Part_I_Fundamental_Physics/`.

DATA AND CODE AVAILABILITY

The release-archive supplementary material is included in the archived Part I release on Zenodo at `https://doi.org/10.5281/zenodo.20250820`. The current version DOI is assigned by Zenodo after each GitHub release. Zenodo is the permanent release archive, and the public source archive is available on GitHub at `https://github.com/bampita-bico/CDFD-Part-I-Release`. The broader public CDFD Runtime is separate reusable software infrastructure; the numerical values reported here are reproduced from this paper's Supplementary Material.

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Author Contributions

Steve Bico Mujjabi: conceptualization, methodology, formal analysis, software, validation, visualization, writing–original draft, and writing–review and editing.

Conflicts of Interest

The author declares no conflicts of interest.

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